Year 11 Higher

REVISION PACK

Algebra 9

ANSWERS

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Algebra 9 · Direct and inverse proportion (Lessons 7.1, 7.2)

Q	Working	Answer
finding the formula, then using it backwards		
1.1	**(a)** $m = kr^3$ $24 = k \times 8$ $k = 3$ **(b)** $81 = 3r^3$ $r^3 = 27$ $r = 3$	**(a)** $m = 3r^3$ **(b)** $r = 3$
1.2	**(a)** $m = kr^3$ $54 = k \times 27$ $k = 2$ **(b)** $128 = 2r^3$ $r^3 = 64$ $r = 4$	**(a)** $m = 2r^3$ **(b)** $r = 4$
1.3	**(a)** $m = kr^2$ $80 = k \times 16$ $k = 5$ **(b)** $125 = 5r^2$ $r^2 = 25$ $r = 5$	**(a)** $m = 5r^2$ **(b)** $r = 5$
1.4	**(a)** $m = kr^3$ $16 = k \times 8$ $k = 2$ **(b)** $250 = 2r^3$ $r^3 = 125$ $r = 5$	**(a)** $m = 2r^3$ **(b)** $r = 5$

1.5	**(a)** $m = kr^3$ $320 = k \times 64$ $k = 5$ **(b)** $135 = 5r^3$ $r^3 = 27$ $r = 3$	**(a)** $m = 5r^3$ **(b)** $r = 3$
1.6	**(a)** $m = kr^2$ $36 = k \times 9$ $k = 4$ **(b)** $100 = 4r^2$ $r^2 = 25$ $r = 5$	**(a)** $m = 4r^2$ **(b)** $r = 5$
inverse proportion to x²		
1.7	$y = \frac{k}{x^2}$ $8 = \frac{k}{9}$ $k = 72$ $y = \frac{72}{36} = 2$	$y = 2$
1.8	$y = \frac{k}{x^2}$ $20 = \frac{k}{4}$ $k = 80$ $y = \frac{80}{16} = 5$	$y = 5$
1.9	$y = \frac{k}{x^2}$ $4 = \frac{k}{25}$ $k = 100$ $y = \frac{100}{4} = 25$	$y = 25$

1.10	$y = \frac{k}{x^2}$ $5 = \frac{k}{16}$ $k = 80$ $y = \frac{80}{4} = 20$	$y = 20$
1.11	$y = \frac{k}{x^2}$ $36 = \frac{k}{1}$ $k = 36$ $y = \frac{36}{9} = 4$	$y = 4$
1.12	$y = \frac{k}{x^2}$ $1.5 = \frac{k}{100}$ $k = 150$ $y = \frac{150}{25} = 6$	$y = 6$
direct proportion to √x		
1.13	$y = k\sqrt{x}$ $10 = k \times 2$ $k = 5$ $y = 5 \times 6 = 30$	$y = 30$
1.14	$y = k\sqrt{x}$ $15 = k \times 5$ $k = 3$ $y = 3 \times 8 = 24$	$y = 24$
1.15	$y = k\sqrt{x}$ $6 = k \times 3$ $k = 2$ $y = 2 \times 9 = 18$	$y = 18$
1.16	$y = k\sqrt{x}$ $8 = k \times 4$ $k = 2$ $y = 2 \times 10 = 20$	$y = 20$

1.17	$y = k\sqrt{x}$ $14 = k \times 2$ $k = 7$ $y = 7 \times 3 = 21$	$y = 21$
1.18	$y = k\sqrt{x}$ $30 = k \times 6$ $k = 5$ $y = 5 \times 2 = 10$	$y = 10$
what happens to y?		
1.19	$3^2 = 9$	y **is multiplied by 9**
1.20	$2^2 = 4$	y **is divided by 4**
1.21	$2^3 = 8$	y **is multiplied by 8**
1.22	x is multiplied by 5	y **is divided by 5**
1.23	$\sqrt{4} = 2$	y **is multiplied by 2**
1.24	$10^2 = 100$	y **is divided by 100**